

Circuit Simulation Lab:17

Design of Synchronous /Asynchronous T Flops

The purpose of this experiment is to introduce the design of Synchronous/asynchronous sequential circuits, in this case T Flipflop.

Code:

Design one circuit which has Synchronous and Asynchronous outputs for a T Flip Flop with Clear and Reset Signals in behavioral modeling and capture waveforms.

```
module tff_sync_async(
input clk,
input t,
input arst,
input srst,
output reg q,
output qbar
);

always @(posedge clk or posedge arst)
begin

if(arst)
    q <= 1'b0;

else if(srst)
    q <= 1'b0;

else if(t)
    q <= ~q;

else
    q <= q;

end

assign qbar = ~q;

Endmodule
```

TESTBENCH

```
module tb_tff_sync_async;
```

```
reg clk;
```

```
reg t;
```

```
reg arst;
```

```
reg srst;
```

```
wire q;
```

```
wire qbar;
```

```
tff_sync_async dut(
```

```
.clk(clk),
```

```
.t(t),
```

```
.arst(arst),
```

```
.srst(srst),
```

```
.q(q),
```

```
.qbar(qbar)
```

```
);
```

```
always #10 clk = ~clk;
```

```
initial begin
```

```
$dumpfile("tff_sync_async.vcd");
```

```
$dumpvars(0,tb_tff_sync_async);
```

```
clk = 0;
```

```
t = 0;
```

```
arst = 0;
```

```
srst = 0;
```

```
// Toggle
```

```
#15;
```

```
t = 1;
```

```
#20;
```

```
// Toggle again
```

```
#20;
```

```
// Asynchronous Reset
```

```
arst = 1;
```

```
#5;

arst = 0;

// Toggle
#20;
t = 1;

#20;

// Synchronous Reset
srst = 1;

#5;

// Wait for next posedge clk
#20;

srst = 0;

#20;
t = 1;

#40;

$finish;

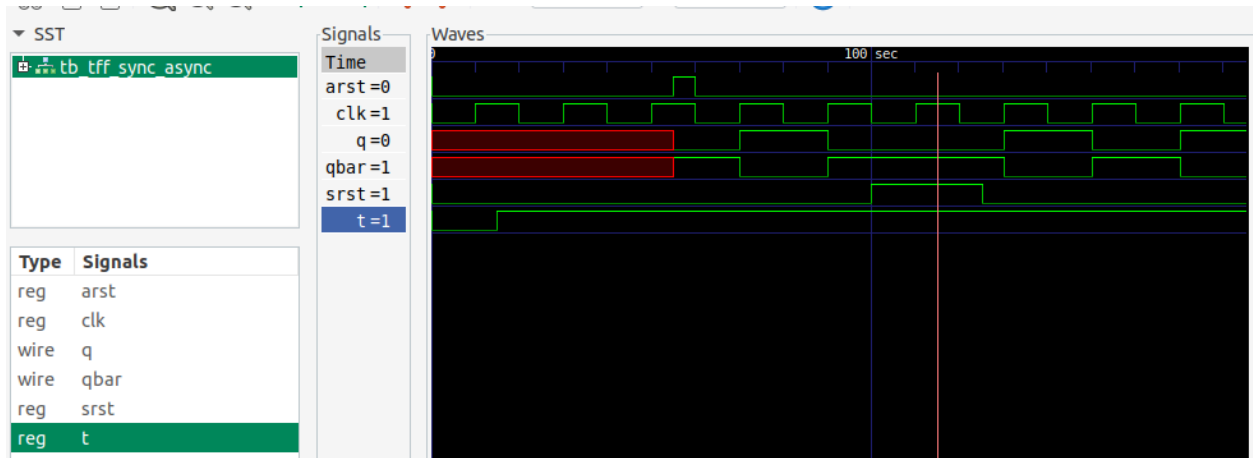
end

initial begin

$monitor(
"time=%0t clk=%b t=%b arst=%b srst=%b q=%b qbar=%b",
$time,clk,t,arst,srst,q,qbar
);

end

Endmodule
```



```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_17$ iverilog -o out tff_sync_async.v tb_tff_sync_async.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_17$ vvp out
VCD info: dumpfile tff_sync_async.vcd opened for output.
time=0 clk=0 t=0 arst=0 srst=0 q=x qbar=x
time=10 clk=1 t=0 arst=0 srst=0 q=x qbar=x
time=15 clk=1 t=1 arst=0 srst=0 q=x qbar=x
time=20 clk=0 t=1 arst=0 srst=0 q=x qbar=x
time=30 clk=1 t=1 arst=0 srst=0 q=x qbar=x
time=40 clk=0 t=1 arst=0 srst=0 q=x qbar=x
time=50 clk=1 t=1 arst=0 srst=0 q=x qbar=x
time=55 clk=1 t=1 arst=1 srst=0 q=0 qbar=1
time=60 clk=0 t=1 arst=0 srst=0 q=0 qbar=1
time=70 clk=1 t=1 arst=0 srst=0 q=1 qbar=0
time=80 clk=0 t=1 arst=0 srst=0 q=1 qbar=0
time=90 clk=1 t=1 arst=0 srst=0 q=0 qbar=1
time=100 clk=0 t=1 arst=0 srst=1 q=0 qbar=1
time=110 clk=1 t=1 arst=0 srst=1 q=0 qbar=1
time=120 clk=0 t=1 arst=0 srst=1 q=0 qbar=1
time=125 clk=0 t=1 arst=0 srst=0 q=0 qbar=1
time=130 clk=1 t=1 arst=0 srst=0 q=1 qbar=0
time=140 clk=0 t=1 arst=0 srst=0 q=1 qbar=0
time=150 clk=1 t=1 arst=0 srst=0 q=0 qbar=1
time=160 clk=0 t=1 arst=0 srst=0 q=0 qbar=1
time=170 clk=1 t=1 arst=0 srst=0 q=1 qbar=0
time=180 clk=0 t=1 arst=0 srst=0 q=1 qbar=0
```